Kehan Guo

kguo2@nd.edu kehanguo2.github.io github.com/KehanGuo2 Google Scholar

SUMMARY

My research focuses on **RL post-training and reward design for LLM agents**, organized around one question: when a model is optimized against a proxy — a reward model, a verifier, a frozen environment — where does the optimization pressure go? I work across three threads:

- **Reward design and RL post-training:** reward-aligned generation through a noise-space alignment interface [7], causally-enhanced policy optimization [10], and adaptive inference-time reasoning [2].
- **Agentic systems and their environments:** how verifiers, tools, and corpora shape — and are exploited by — the policies trained against them: tool-orchestrating scientific agents [3], runtime enforcement for coding agents [8], and live agentic evaluation [9].
- **Rigorous scientific evaluation:** benchmarks for how models acquire, verify, and act on information [1, 4, 5], and generative modeling for scientific discovery [6].

EDUCATION

University of Notre Dame Aug 2022 – May 2027 (expected)
Ph.D. in Computer Science · Advisor: Prof. Xiangliang Zhang (MINE Lab)

Boston University 2020 – 2022
M.S. in Computer Science

RESEARCH EXPERIENCE

Apple Jun 2026 – present
Research Intern · Cupertino, CA

- Research area: reinforcement-learning post-training for large language models.

ByteDance Apr 2026 – Jun 2026
Research Scientist Intern · San Jose, CA

- Built BI-Claw, an internal tool-using LLM agent for business-integrity risk investigation, orchestrating 20+ domain skills over internal data services.
- Designed a self-evolving skill pipeline — skills seeded from static evaluation, evolved from real user queries with LLM-as-judge feedback — raising task success from 62% to 70% on an internal benchmark.

Amazon AWS AI May 2025 – Aug 2025
Applied Scientist Intern · New York, NY

- Designed **Social-Choice GRPO**, a reward shaping framework that converts sparse binary RL signals into dense rank-based feedback via social choice aggregation across policy rollouts, reducing gradient variance on long-horizon reasoning chains.
- Built a model-agnostic critique layer fusing rule-based verification with calibrated model confidence, enabling stable policy gradient updates on ambiguous, multi-step reasoning paths.
- Improved accuracy and sample efficiency over PPO baselines across math, coding, and logical reasoning tasks.

University of Notre Dame — MINE Lab Aug 2022 – Present
Research Assistant · AI for Science

- Building toward a **reliable scientific-discovery system** a domain scientist can trust end to end — grounded in the rigorous chemistry benchmarks I created [1, 4] and extended to agents that retrieve, verify, and reason over real evidence.

- ChemKG (in submission): a chemistry knowledge graph with an agentic natural-language engine that grounds every answer — reaction retrieval, condition recommendation, and hypothesis generation — in real reaction records, targeting medicinal-chemistry workflows.

SELECTED PUBLICATIONS

* equal contribution. Full list at Google Scholar.

[1] Can LLMs Solve Molecule Puzzles? A Multimodal Benchmark for Molecular Structure Elucidation.
K. Guo, B. Nan, Y. Zhou, et al. *NeurIPS 2024* **[Spotlight]**

[2] AdaReasoner: Adaptive Reasoning Enables More Flexible Thinking.
X. Wang, Y. Huang, Y. Wang, X. Luo, **K. Guo**, Y. Zhou, X. Zhang. *NeurIPS 2025* **[Spotlight]**

[3] ChemOrch: Empowering LLMs with Chemical Intelligence via Synthetic Instructions.
Y. Huang, Z. Jiang, X. Luo, **K. Guo**, et al. *NeurIPS 2025*

[4] What Can Large Language Models Do in Chemistry? A Comprehensive Benchmark on Eight Tasks.
T. Guo*, **K. Guo***, B. Nan, et al. *NeurIPS 2023*

[5] Evaluating Large Language Models in Scientific Discovery.
Z. Song*, J. Lu*, Y. Du, B. Yu, T. M. Pruyn, Y. Huang, **K. Guo***, et al. *arXiv 2025*

[6] Artificial Intelligence in Spectroscopy: Advancing Chemistry from Prediction to Generation and Beyond.
K. Guo, Y. Shen, et al. *IJCAI 2025* **[Survey Track]**

Preprints / Under Review

[7] Reward Transport: Property Control in Flow Matching via Noise-Space Alignment.
K. Guo, Y. Shen, Y. Zhou, Y. Huang, C. Gao, S. Du, X. Zhang. *Under review* [Code]

[8] Getting Better at Working With You: Compiling User Corrections into Runtime Enforcement for Coding Agents.
Y. Zhou, **K. Guo**, H. Zhuang, et al. *arXiv 2026* [arXiv]

[9] ChemElucid-Live: Live Evaluation of Tool-Using Scientific Agents.
K. Guo, et al. *Under review*

[10] Causally-Enhanced Reinforcement Policy Optimization.
X. Wang, Y. Huang, Y. Zhou, X. Luo, **K. Guo**, X. Zhang. *arXiv 2025*

TECHNICAL SKILLS

Python, PyTorch, JAX, Megatron-LM, DeepSpeed ZeRO, VeRL, Ray, GRPO/PPO/DPO, reward modeling, vLLM, HF Transformers, LLM agents, tool use, RAG, Claude Code

AWARDS & RECOGNITION

NeurIPS 2024 Spotlight — *MolPuzzle* (first author)
NeurIPS 2025 Spotlight — *AdaReasoner* (co-author)
OpenAI Researcher Access Program (2024)

PROFESSIONAL SERVICE

Reviewer: NeurIPS (2024–25), ICLR (2025–26), ICML, AAAI, IJCAI, KDD, WWW, ACL Rolling Review (ACL/EMNLP)
Tutorial: From VAEs to Diffusion and LLMs: Modern Generative Models for Molecular Discovery, KDD 2026
Organizer: Workshop on Reliable Scientific Foundation Models (RelSciFM), KDD 2026